

Lab 17: Exam 03 Review — Modules 12–15

BIOL-1

Name: _____ Date: _____

Purpose

This lab is **practice** for the third unit: **evolution in populations, speciation, and ecology**. It pulls together ideas from your Keys to Success and class notes. It is **not** a copy of the official exam; wording and topics are reordered and rephrased so you practice **concepts**, not one memorized set of sentences.

How to use this lab

1. Try the multiple-choice and word-bank items **closed-book** first, then reopen notes only where you were stuck.
2. For **short answers**, write in full sentences (or labeled bullet lists where that matches your instructor's expectations).
3. Ask your instructor for the **official exam** instructions, timing, and point breakdown—those rules override anything informal here.

Module 12 — Darwin and evolution

Focus ideas

- **Descent with modification** (change over time from shared ancestors).
- **Natural selection**: heritable **variation**, **overproduction** of offspring, **differential reproductive success** (fitness = relative reproductive success).
- **Individuals** do not evolve; **populations** change in allele frequency over generations.
- **Evidence** from fossils, biogeography, comparative anatomy (homology), molecular comparisons.

Quick drill — write the letter (A–D)

1. In biology, **evolutionary fitness** is best defined as:
A) Physical strength relative to other species

- B) Survival time without food
- C) Relative reproductive success (passing alleles to the next generation)
- D) Ability to acclimate to cold weather in one lifetime

2. Which statement best matches "**descent with modification**"?

- A) Offspring are identical to parents in every trait
- B) Species share ancestry and accumulate differences in different environments over time
- C) Individuals intentionally adapt their DNA when the environment changes
- D) Only fossils evolve; living species do not

3. A tall plant and a short plant in the same population both survive drought equally well, but the tall one produces more seeds. Drought is selecting on **height**; the difference in seed number reflects:

- A) Stabilizing selection only
- B) Differential reproductive success despite similar survival here
- C) No role for heritable variation
- D) Acclimation of individuals without genetic change

4. Which is **strong** evidence of **common descent**?

- A) Two species live on different continents, so they must be unrelated
- B) The same **homologous** bone layout in forelimbs of a whale, a bat, and a human, with different functions
- C) Unrelated lineages that look alike because of convergent selection
- D) A single individual that lived 200 years and changed its traits

5. "Survival of the fittest" is often misunderstood. The biologist's "fittest" in this course usually ties most closely to:

- A) Winning fights
- B) Outliving all competitors to age 100
- C) Producing more surviving offspring, on average, than other genotypes in that environment
- D) The largest body size in the population

6. **Populations** evolve, not individuals, because evolution is:

- A) A change in muscle size after weight training
- B) A change in **allele frequencies in a population** across generations
- C) A change in an organism's learned behavior only
- D) A single organism's new mutation expressed in one body cell only

Reflect: In one sentence, name one line of **evidence** for evolution and what it shows.

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Module 13 — How populations evolve (microevolution)

Focus ideas

- **Microevolution** = **change in allele frequencies** in a population over time.
- **Five mechanisms:** mutation, **gene flow**, **genetic drift**, non-random mating (e.g. sexual selection), **natural selection**.
- **Drift** matters most in **small** populations; **bottleneck** vs **founder** effects.
- **Selection modes** on a continuous trait: **directional**, **stabilizing**, **disruptive**.
- **Hardy–Weinberg** = mathematical baseline of **no** evolution; real populations usually violate it.

Quick drill — write the letter (A–D)

1. Which option is a **mechanism of microevolution**?

- A) A sunburn that is not passed to children
- B) **Gene flow** when individuals enter the population and mate
- C) A population count going up because of immigration without allele change
- D) A species inventing a new organ when needed

2. A hurricane randomly kills 90% of a moth population. Surviving moths rebuild the next generation, but many alleles are gone forever. This scenario illustrates:

- A) Stabilizing selection
- B) The **bottleneck** effect (genetic drift)
- C) Sympatric speciation
- D) Density-independent **limit** that is not evolutionary

3. A few lizards are accidentally transported to a new island and found a new population. The island gene pool only reflects the alleles those few carried. This is:

- A) The **founder** effect (genetic drift)
- B) The bottleneck effect in the source population on the mainland
- C) Convergent evolution
- D) Interspecific competition

4. In **directional** selection on body size, the population typically:

- A) Favors the middle of the size range and removes extremes
- B) Splits to favor two opposite extremes and often weakens the middle
- C) Shifts the average toward **one** extreme of the range
- D) Stops all reproduction until Hardy–Weinberg holds

5. In **stabilizing** selection (e.g., human birth weight in some datasets):

- A) The extremes are favored
- B) **Intermediate** phenotypes are favored; extremes are selected against
- C) The population must be on two islands
- D) Only mutations matter; selection does not

6. The Hardy–Weinberg **equilibrium** is useful mainly as:

- A) A proof that all populations stop evolving in nature
- B) A baseline **null** model to compare to real data (infinite size, no drift, no mutation, no migration, random mating, no selection)
- C) A law that only applies to bacteria
- D) A replacement for the biological species concept

Two terms you confuse:

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Module 14 — Macroevolution and speciation

Focus ideas

- **Biological species concept** (interbreed in nature → viable fertile offspring; **reproductive isolation** from other such groups).
- **Prezygotic** barriers (before zygote): habitat, **temporal**, behavioral, mechanical, gametic, etc.
- **Postzygotic** barriers (after zygote): hybrid viability, hybrid fertility, hybrid breakdown.
- **Allopatric** speciation: geographic split then divergence. **Sympatric** speciation: same place (e.g. **polyploidy** in plants).

Quick drill — write the letter (A–D)

1. Two fish species that spawn in the same lake but in different **months** are separated mainly by:

- A) A postzygotic barrier (hybrid sterility)
- B) A **prezygotic** barrier (temporal isolation)
- C) The Hardy–Weinberg theorem
- D) Exponential growth only

2. A horse and a donkey can produce a mule, but the mule is typically sterile. This outcome is:

- A) A prezygotic gametic failure
- B) A **postzygotic** barrier to gene flow (reduced hybrid fertility)
- C) Stabilizing selection in horses
- D) The founder effect

3. A mountain range forms and cuts a **single** species into two populations that no longer exchange genes; over time they diverge until they do not interbreed. This pattern fits:

- A) **Allopatric** speciation
- B) Only sympatric speciation with polyploidy
- C) Macroevolution without microevolution
- D) A species concept based only on appearance

4. In plants, a single generation can create reproductively isolated lineages with extra chromosome sets. That route is a classic example of:

- A) The bottleneck effect only
- B) **Sympatric** speciation via **polyploidy**
- C) A prezygotic mechanical barrier in animals only
- D) Stabilizing selection on flower color

5. Courtship songs differ between two close bird species, so they rarely attempt mating. This isolating mechanism is best described as:

- A) Postzygotic hybrid breakdown
- B) A **prezygotic behavioral** barrier
- C) Exponential population growth
- D) A density-independent limit

6. "Macroevolution" in this course is **not** a separate "kind" of evolution from microevolution; it refers to:

- A) Only fossils, never DNA
- B) Larger-scale results (e.g. new species, major diversity) that still build from the same **population**-level changes over time
- C) Changes that only happen in one generation
- D) Evolution that violates natural selection

In two sentences: Why does **reproductive isolation** allow two lineages to accumulate **different** adaptations over time?

Module 15 — Population and systems ecology

Focus ideas

- **Exponential (J)** vs **logistic (S)** growth; **carrying capacity (K)**.
- **Density-dependent** (competition, disease, predation often intensify with crowding) vs **density-independent** (e.g. many fires, some storms) regulation.
- **Trophic levels** and **~10% energy** transfer; **energy flows, nutrients** cycle.
- **Producers, consumers, decomposers**; food chains limited in length partly because of energy loss as **heat** and work.

Quick drill — write the letter (A–D)

1. A population that starts fast, then the growth **rate** slows as the population nears a stable cap, is best modeled by:

- A) Unlimited exponential growth forever
- B) A **logistic (S) curve** approaching **K**

- C) A straight line of zero growth from year one
- D) A crash to zero in every case

2. A disease spreads more easily when hosts are very crowded. That limiting factor is best classified as:

- A) Density-**independent**
- B) Density-**dependent**
- C) Only abiotic (never biological)
- D) Not part of population ecology

3. A flash flood kills a similar **fraction** of a rare plant whether the local patch has 20 or 2,000 individuals. (Treat the event as a regulator that does not depend on **per-capita** crowding the way disease might.) The flood acts mainly as:

- A) A density-dependent **biotic** limit only
- B) A **density-independent** disturbance (still severe for the population)
- C) Proof that K does not exist
- D) Stabilizing selection on roots

4. The **~10% rule** between trophic levels implies that, compared with eating **plants** directly, a human food system that relies on many steps of long food chains to **tuna** would require:

- A) **More** net primary production to deliver the same food calories to people
- B) **Less** primary production because fish are more efficient than plants
- C) The same amount of energy at every level
- D) Decomposers to create new energy from nothing

5. In ecosystems, **chemical elements** (e.g. nitrogen, carbon in nutrients) are **cycled**, while **energy** is:

- A) Cycled the same way as nitrogen
- B) **Lost as heat** along the way; not recycled to be photosynthesized again from dead matter in the same way sunlight is new input
- C) Only stored in apex predators
- D) Only found in the atmosphere, never in plants

6. Decomposers (fungi, bacteria) are critical because they:

- A) Add new energy from the sun to dead matter
- B) **Break down** organic material and return **nutrients** to forms producers can use again
- C) Make trophic levels unnecessary
- D) Only affect density-dependent limits

Connect: In one sentence, link **K** to **energy** (why many organisms at the top need a big base of producers).

Word bank sprint

Use each term once. Practice meaning, not one memorized sentence shape.

Word Bank: *allele frequencies, allopatric, biological species, density-dependent, descent with modification, decomposers, directional selection, homologous, prezygotic, stabilizing selection*

1. In formal terms, **microevolution** means a change in _____ from one generation to the next in a population.
2. In comparative anatomy, forelimb bones in different tetrapods that share structure from a common ancestor are called _____ structures (even if functions differ).
3. A population's birth-weight curve can favor the middle; this mode of natural selection is _____ selection.
4. A barrier that works **before** a zygote forms—such as different breeding seasons—is a _____ barrier to reproduction.
5. A river splits a range; populations diverge in isolation—speciation that begins with **geography** is often _____ speciation.
6. Darwin's phrase for evolution: species share ancestry and accrue change over time—_____ with modification.
7. A limiting factor that hits harder as a population **crowds** (e.g. more transmission of a parasite) is _____ regulation in many textbook examples.
8. A group of populations that interbreed in nature, producing viable fertile offspring, and is reproductively isolated from other such groups, fits the _____ species concept (as presented in this unit).
9. Fungi and bacteria that recycle dead organic matter and return elements to the soil for producers are _____.

10. When a population's average trait value shifts toward **one** extreme of the range (e.g. darker moths on sooty trees), the pattern is _____ selection.

Short-answer rehearsal

Draft at least three in your own words. On your exam, you may answer a subset of longer prompts check your real instructions.

1. (Module 12) Contrast how **individuals** can change in a lifetime with how **evolution** is defined in this course. Name **one** type of **evidence** for evolution in one sentence.

2. (Module 13) Compare **genetic drift** to **natural selection** (what drives change in each?). Give **one** example of **bottleneck** or **founder** effect in plain words.

3. (Module 14) Classify these as **prezygotic** or **postzygotic**: (a) different courtship song; (b) hybrid is sterile. Then state one difference between **allopatric** and **sympatric** speciation in one or two sentences.

4. (Module 15) Draw the distinction between **density-dependent** and **density-independent** limits, then use the **~10% rule** to explain why long food chains support **fewer** high-level consumers than a base of the same size could support if humans ate **primary producers** directly. (Equations optional; clear logic is enough.)

Self-check before the exam

I can...	✓
State natural selection as variation + overproduction + unequal contribution to the next generation	[]
Give one type of evidence (fossil, anatomy, biogeography, molecular) and what it supports	[]
Name the five microevolution mechanisms; contrast drift vs gene flow	[]
Describe directional / stabilizing / disruptive selection; explain bottleneck vs founder	[]
Classify prezygotic vs postzygotic; allopatric vs sympatric; biological species idea	[]
Contrast J vs S growth; K; density-dependent vs independent; energy flow vs nutrient cycling ; 10% rule	[]

Two terms I still mix up:

One action before the exam (be specific):